

Convolutional Neural Network (CNN)/convnet

CNN are special kind of Neural Network for processing data that has known grid like topology like time series data (1D) or images (2D)

ANN $\rightarrow$ Matrix Multiplication

CNN $\rightarrow$ Convolutional operation

image $\rightarrow$ pixel has 2D grid

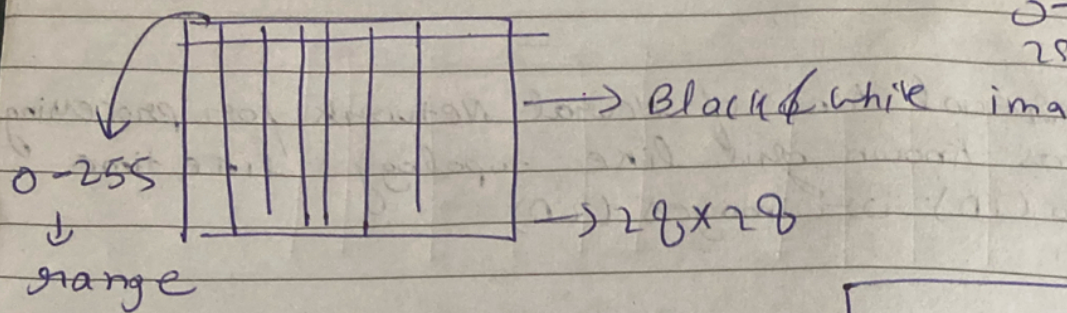
ANN can work for images data but we use CNN because they are effective because ANN

- 1) High computation cost
- 2) overfitting
- 3) Loss of imp info
like spatial arrangement
of pixels

CNN VS visual cortex

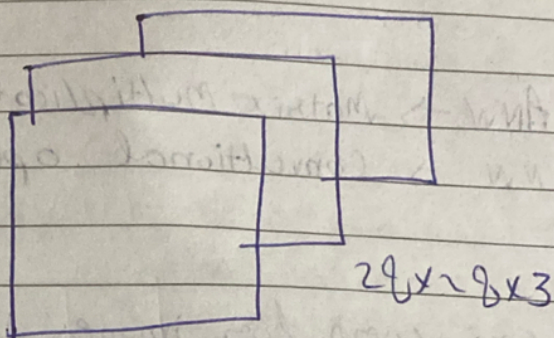
CNN is ideology drawn from visual part of human body

Convolutional operation



0 → Black
255 → white

for colour images



3 layers for RGB

How to detect Edge

→ The intensity of pixel is changed

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
255	255	255	255	255	255
255	255	255	255	255	255
255	255	255	255	255	255

$n \times m$

-1	1	-1
0	0	0
1	1	1

filter → kernel
 $m \times m$

$(n-m+1) \times (m-m+1)$

Steps

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
255	255	255	255	255	255
11	11	11	11	11	11
11	11	11	11	11	11

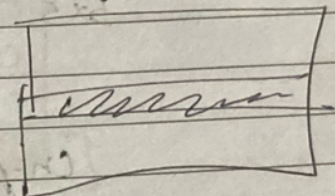
-1	-1	-1
0	0	0
1	1	1

horizontal filter

0	0	0	0
255	255	255	255
255	255	255	255
11	11	11	11

-1	0	1
-1	0	1
-1	0	1

vertical filter



for RGB filter will be $3 \times 3 \times 3$
each for each layer